PHOTOSYNTHESIS: HOW PLANTS MAKE FOOD
7th Grade Life Science -- Handout 1

Overview

Photosynthesis is the process plants, algae, and some bacteria use to convert light energy into chemical energy stored in sugar. It is the foundation of nearly every food chain on Earth: without it, animals (including us) would have no food to eat and no oxygen to breathe.

The Word Equation

$$6\,CO_2 + 6\,H_2O \xrightarrow{\text{light energy}} C_6H_{12}O_6 + 6\,O_2$$

(carbon dioxide + water) (glucose) (oxygen)

In plain English: a plant pulls in carbon dioxide from the air and water from the soil. Using sunlight as the energy source, it rearranges those atoms into a sugar molecule (glucose) and releases oxygen as a waste product. The plant uses the sugar for growth, repair, and to make other molecules such as starch, cellulose, fats, and proteins.

Where It Happens

Photosynthesis takes place mostly in the leaves, inside small green structures called CHLOROPLASTS. Chloroplasts contain a pigment called CHLOROPHYLL, which absorbs red and blue light strongly and reflects green light -- that is why leaves look green to our eyes.

A leaf is built like a tiny solar factory:

- The waxy upper surface (cuticle) limits water loss.
- Pores on the underside (stomata) let CO_2 in and O_2 out.
- Veins carry water up from the roots and sugar back to the rest of the plant.

The Two Stages

Photosynthesis is really two linked sets of reactions.

1) Light-dependent reactions (in the thylakoid membranes): Sunlight splits water molecules. This releases oxygen, produces energy carriers called ATP and NADPH, and is the only step that directly uses light.

2) Light-independent reactions, also called the Calvin cycle (in the stroma): The ATP and NADPH from stage 1 power a series of reactions that "fix" CO_2 into sugar. These reactions do not need light directly, but they need the products of stage 1, so

they slow down at night.

Factors That Change the Rate

The rate of photosynthesis -- how fast a plant makes sugar -- depends on three main factors:

* LIGHT INTENSITY. More light usually means more photosynthesis, up to a saturation point where the chloroplasts can't keep up.
* CARBON DIOXIDE CONCENTRATION. Higher CO_2 in the air boosts the rate until other factors become limiting. Greenhouse growers sometimes enrich CO_2 for this reason.
* TEMPERATURE. The enzymes that run the Calvin cycle work best in a middle range (roughly 20-35 degrees C for most plants). Too cold and they slow down; too hot and they denature.

Water availability also matters: when a plant runs short on water it closes its stomata to conserve moisture, but that also cuts off the CO_2 supply, so photosynthesis stalls.

Why It Matters

Photosynthesis is the link between the sun and almost every living thing.

- The OXYGEN we breathe comes from photosynthesis in plants, algae, and cyanobacteria.
- The FOOD ENERGY in everything we eat -- fruit, bread, beef, fish -- can be traced back to a plant or alga that captured sunlight.
- The CARBON CYCLE depends on photosynthesis pulling CO_2 out of the atmosphere; this is why forests and oceans are called "carbon sinks."

Key Vocabulary

chloroplast .... organelle where photosynthesis takes place
chlorophyll .... green pigment that absorbs light
stomata .... pores on a leaf that let gases in and out
glucose .... simple sugar produced by photosynthesis
Calvin cycle ... light-independent stage that builds sugar from CO_2
ATP / NADPH .... energy carriers made in the light-dependent stage

Quick Check

1. Where in the plant cell does photosynthesis occur?
2. What two raw materials does a plant combine to make glucose?
3. Why do leaves look green?
4. Name two factors that can speed up or slow down the rate of photosynthesis, and explain why each one matters.

5. In one sentence, describe how the energy stored in a hamburger originally came from the sun.